Configuration Still Beats Algorithm, Until the Drift Hides in the Correlations

Extending Resource-Constrained Drift Detection to Multivariate Shift

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Motivation: Two Open Questions from Paper 1

Paper 1's finding

Monitoring configuration (frequency, window size) beats detector algorithm choice, by a 37-fold margin in detection delay, for three univariate detectors.

Two questions that finding leaves open:

1. Does this survive extending to **genuinely multivariate** detectors?
2. Rabanser et al. (2019, NeurIPS) found univariate testing **rivals** multivariate testing, on shifts that all move some feature's own distribution. What if a shift leaves **every marginal untouched**?

This study

Two experimental grids, one that extends Paper 1's condition to multivariate detectors, one built specifically so univariate testing *cannot* succeed.

Methodology

Same 3 datasets, now with 2 correlated features each

- Credit: AGE + BILL_AMT1
- Telco: Monthly + Total Charges
- Maintenance: Tool wear + Air temp.

6 detectors

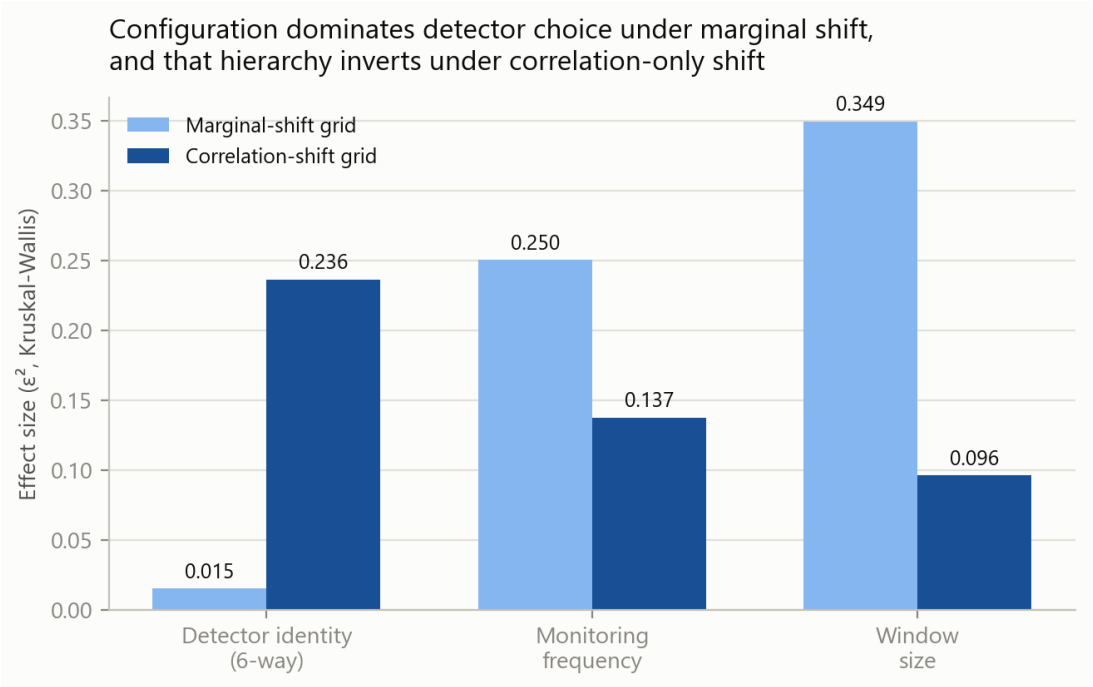
- Univariate: KS, PSI, Wasserstein
- Multivariate: MMD, C2ST, HDDDM

Two drift conditions

- **Marginal**: shifts each feature's own distribution (16,200 runs)
- **Correlation-only**: rank-permutation, marginals provably unchanged (3,240 runs)

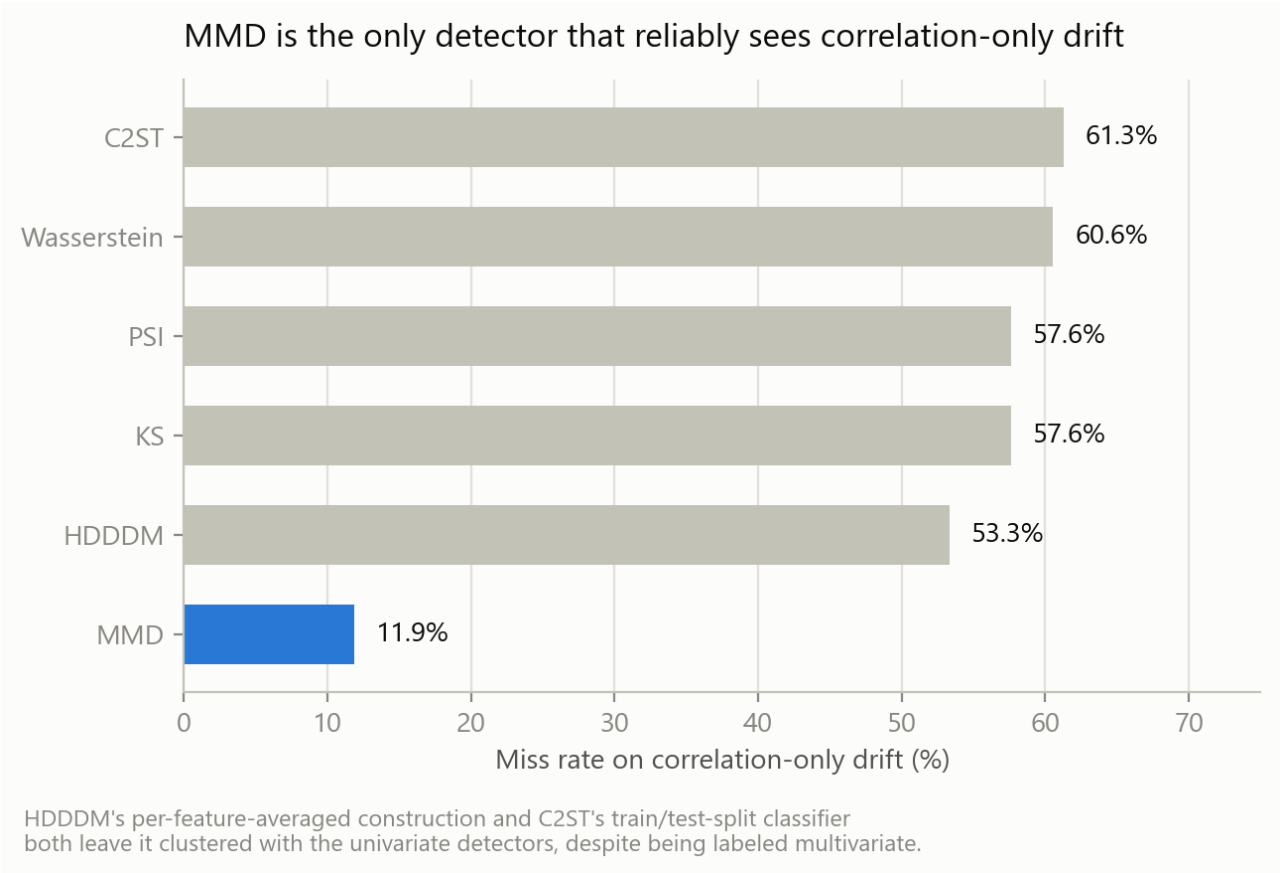
Kruskal-Wallis, Mann-Whitney U, Spearman ρ, χ^2, study-wide FDR correction

Result 1: Configuration Dominance Replicates



- Marginal grid: window size $\varepsilon^2 = 0.349$ vs. detector identity $\varepsilon^2 = 0.015$
- Detector **class** has *no* effect on whether drift is caught at all (χ^2, $p = 0.21$)
- Paper 1's finding holds under a harder, multivariate setting

Result 2: The Hierarchy Inverts, Sharply



- Correlation-only grid: detector identity $\varepsilon^2 = 0.236$, now **larger** than either constraint effect
- Whether drift is caught at all: $\chi^2 = 86.7$, $p = 1.3 \times 10^{-20}$

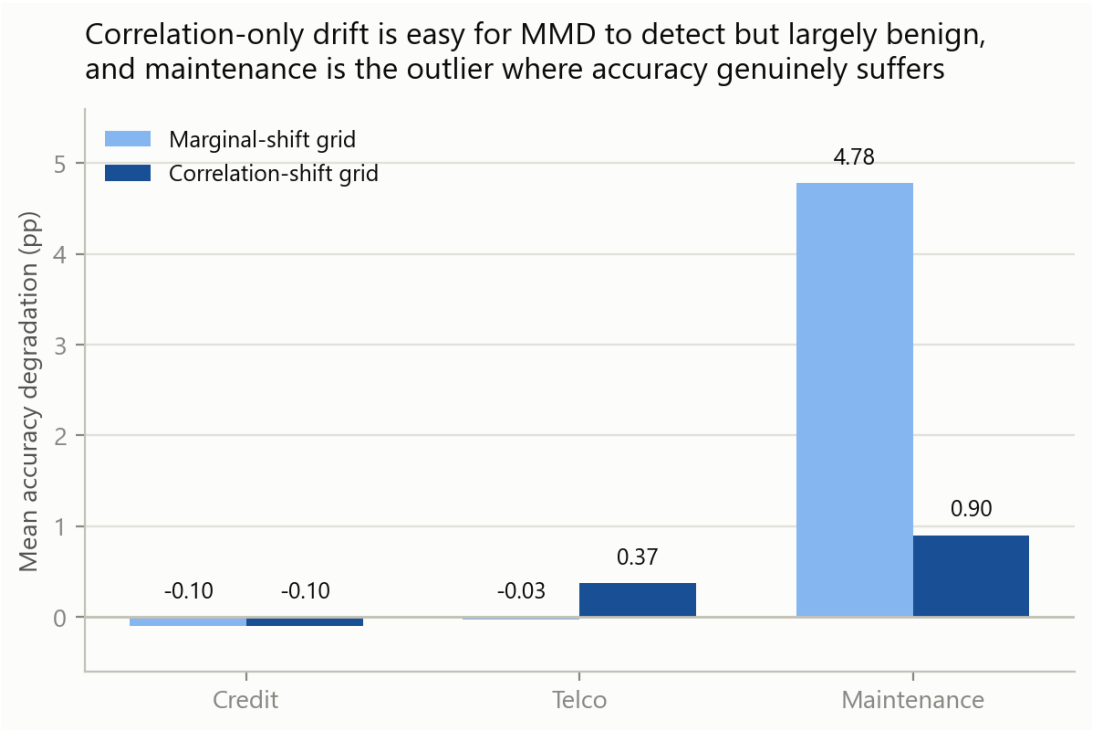
The Real Story: Construction, Not Class

Detector	Miss rate	Sees joint structure?
MMD	11.9%	Yes, by kernel construction
HDDDM	53.3%	**No**: averages per-feature distances
C2ST	61.3%	In principle, but sample-starved

The point

HDDDM is labelled multivariate but its own published formula (Ditzler & Polikar, 2011) mathematically cannot see correlation structure. The label doesn't predict performance; the statistic's construction does.

Correlation-Only Drift: Easy to Catch, Mostly Benign



- Maintenance is the outlier where accuracy genuinely suffers
- Credit and Telco: near-zero impact in both conditions
- Echoes Rabanser et al.'s distinction: detectable $\neq$ malignant

Takeaways for Practitioners

1. **Configuration first**, still, whenever drift has a marginal signature
2. If monitored features could drift **jointly** without moving individually: detector choice now matters most
3. Don't trust the **multivariate label** alone: check whether the statistic actually uses cross-feature dependency

Limitations

Two-feature pairs only; single model (Random Forest); Wasserstein threshold empirically, not literature, calibrated; synthetic injection throughout

Next steps

Higher-dimensional feature groups; C2ST at larger windows; embedding-space extension (text/vision) as future work, not required here